



Lab 03: VPC Networks - Controlling Access

Build a Secure Google Cloud Network | GSP213

Overview

Created two nginx web servers on the default VPC and controlled external HTTP access using tagged firewall rules. Then explored IAM Network Admin and Security Admin roles using a service account to understand permission boundaries.

Objectives

- Create nginx web servers with and without network tags
- Create a tagged firewall rule to control HTTP access
- Create a service account with Network Admin role
- Test and compare Network Admin vs Security Admin permissions

Task 1 - Create Web Servers

Created two nginx web servers: blue (with web-server tag) and green (without tag). Installed nginx on both and customized their home pages.

- blue server: has web-server network tag
- green server: no network tag

```
sudo apt-get install nginx-light -y
sudo nano /var/www/html/index.nginx-debian.html
```

The screenshot shows the Google Cloud Platform 'VM instances' page. At the top, there are tabs for 'Instances', 'Observability', and 'Instance schedules'. Below the tabs, there's a search bar and a table of VM instances. The table has columns for Status, Name, Zone, Recommendations, In use by, Internal IP, External IP, and Connect. Two instances are listed: 'blue' and 'green', both in the 'europe-west1-b' zone. The 'blue' instance has an internal IP of 10.132.0.2 and an external IP of 35.205.7.35. The 'green' instance has an internal IP of 10.132.0.3 and an external IP of 207.175.91.151. Below the table, there's a 'Related actions' section with icons for various actions like 'Create instance', 'Import VM', and 'Refresh'.

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☐	blue	europe-west1-b			10.132.0.2 (nic0)	35.205.7.35 (nic0)	SSH
☐	green	europe-west1-b			10.132.0.3 (nic0)	207.175.91.151 (nic0)	SSH

Task 2 - Create the Firewall Rule

Created a tagged firewall rule that only allows HTTP traffic to VMs with the web-server tag.
Created a test-vm to verify access.

- Rule name: allow-http-web-server
- Target tags: web-server
- Protocol: TCP port 80 and ICMP
- Source: 0.0.0.0/0

```
gcloud compute instances create test-vm --machine-type=e2-micro --subnet=default --zone=[ZONE]
```

HTTP connectivity results:

- blue internal IP: WORKED (same VPC default-allow-internal rule)
- green internal IP: WORKED (same VPC default-allow-internal rule)
- blue external IP: WORKED (has web-server tag)
- green external IP: FAILED (no web-server tag)

```
student_01_af3616165307@blue:~$ sudo nano /var/www/html/index.nginx-debian.html
student_01_af3616165307@blue:~$ cat /var/www/html/index.nginx-debian.html
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to the blue server!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
student_01_af3616165307@blue:~$
```

```
ssh.cloud.google.com/v2/ssh/projects/qwiklabs-gcp-01-2b20054
SSH-in-browser

Preparing to unpack .../libnginx-mod-http-echo_1%3a0.63
Unpacking libnginx-mod-http-echo (1:0.63-4) ...
Selecting previously unselected package nginx-light.
Preparing to unpack .../nginx-light_1.22.1-9+deb12u7_al
Unpacking nginx-light (1.22.1-9+deb12u7) ...
Setting up nginx-common (1.22.1-9+deb12u7) ...
Created symlink /etc/systemd/system/multi-user.target.w
Setting up nginx (1.22.1-9+deb12u7) ...
Upgrading binary: nginx.
Setting up libnginx-mod-http-echo (1:0.63-4) ...
Setting up nginx-light (1.22.1-9+deb12u7) ...
Processing triggers for man-db (2.11.2-2) ...
Processing triggers for nginx (1.22.1-9+deb12u7) ...
Triggering nginx reload ...
student_01_af3616165307@green:~$ sudo nano /var/www/html
student_01_af3616165307@green:~$ cat /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to the green server!</h1>
<p>If you see this page, the nginx web server is succes
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
student_01_af3616165307@green:~$
```

Task 3 - Explore Network and Security Admin Roles

Permission	Network Admin	Security Admin
List firewall rules	✓ Yes	✓ Yes
Delete firewall rules	✗ No	✓ Yes
Modify network resources	✓ Yes	✗ No

```
gcloud auth activate-service-account --key-file credentials.json
gcloud compute firewall-rules list
gcloud compute firewall-rules delete allow-http-web-server
```

```
SSH-in-browser
student_01_af3616165307@test-vm:~$ gcloud auth activate-service-account --key-file credentials.json
Activated Service account credentials for: [network-admin@qwiklabs-gcp-01-2b20054fa23f.iam.gserviceaccount.com]
student_01_af3616165307@test-vm:~$ gcloud compute firewall-rules list
NAME_ NETWORK_ DIRECTION_ PRIORITY_ ALLOW_ DENY_ DISABLED
allow-http-web-server default INGRESS 1000 tcp:80,icmp False
default-allow-icmp default INGRESS 65534 icmp False
default-allow-internal default INGRESS 65534 tcp:0-65535,udp:0-65535,icmp False
default-allow-rdp default INGRESS 65534 tcp:3389 False
default-allow-ssh default INGRESS 65534 tcp:22 False

To show all fields of the firewall, please show in JSON format: --format=json
To show all fields in table format, please see the examples in --help.

student_01_af3616165307@test-vm:~$ gcloud compute firewall-rules delete allow-http-web-server
The following firewalls will be deleted:
- [allow-http-web-server]

Do you want to continue (Y/n)? Y

ERROR: (gcloud.compute.firewall-rules.delete) Could not fetch resource:
- Required 'compute.firewalls.delete' permission for 'projects/qwiklabs-gcp-01-2b20054fa23f/global/firewalls/allow-http-web-server'
```

```
student_01_af3616165307@test-vm:~$ gcloud compute firewall-rules list
NAME_ NETWORK_ DIRECTION_ PRIORITY_ ALLOW_ DENY_ DISABLED
allow-http-web-server default INGRESS 1000 tcp:80,icmp False
default-allow-icmp default INGRESS 65534 icmp False
default-allow-internal default INGRESS 65534 tcp:0-65535,udp:0-65535,icmp False
default-allow-rdp default INGRESS 65534 tcp:3389 False
default-allow-ssh default INGRESS 65534 tcp:22 False

To show all fields of the firewall, please show in JSON format: --format=json
To show all fields in table format, please see the examples in --help.

student_01_af3616165307@test-vm:~$ gcloud compute firewall-rules delete allow-http-web-server
The following firewalls will be deleted:
- [allow-http-web-server]

Do you want to continue (Y/n)? Y

Deleted [https://www.googleapis.com/compute/v1/projects/qwiklabs-gcp-01-2b20054fa23f/global/firewalls/allow-http-web-server].
student_01_af3616165307@test-vm:~$
```

What I Learned

- Network tags allow firewall rules to target specific VMs instead of all VMs
- Tagged firewall rules are more secure than rules that apply to all instances
- Network Admin can manage network resources but cannot modify firewall rules
- Security Admin can create, modify, and delete firewall rules
- Service accounts can be granted specific IAM roles to limit blast radius
- Principle of least privilege means only granting permissions actually needed

Resources

GCP VPC Firewall Docs: <https://cloud.google.com/vpc/docs/firewalls>
Lab ID: GSP213